



Course : R-VIJETA (R-JP)

PHYSICS DPP

DAILY PRACTICE PROBLEM

REVISION DPP NO. # 03

Target : JEE (Advanced) 2023

DPP Syllabus :

ELECTROMAGNETIC FIELD, ELECTROMAGNETIC INDUCTION AND ALTERNATING CURRENT
DPP No. : 03

Total Marks : 170

Max. Time : 160

One or More than one may be correct option ('-1' Negative marking) Q.1 to 20;

(3 Marks; 2 Min.)

[60, 40]

Subjective Questions ('-1' Negative marking) Q.21 to Q.40;

(4 Marks; 5 Min.)

[80, 100]

Single Choice Objective ('-1' Negative marking) Q.41 to Q.50

(3 Marks; 2 Min.)

[30, 20]

1. In a series L, C, R circuit with an ac source, $V_R = 4\text{volt}$, $V_L = 3\text{volt}$ and $V_C = 4\text{volt}$ and $I = 2\text{ amp}$. Mark the correct options, assuming that the values given are rms values.

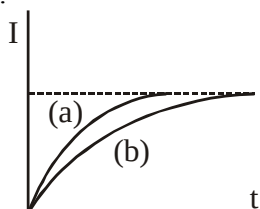
(A) The power factor of the circuit is $\frac{4}{\sqrt{17}}$

(B) The average power consumption is 8 Watts.

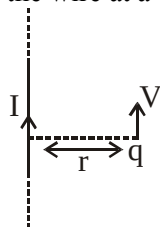
(C) The reactance of the circuit is $1\ \Omega$

(D) The impedance of the circuit is $\frac{\sqrt{17}}{2}\ \Omega$.

2. A circuit consisting of a constant e.m.f. 'E', a self induction 'L' and a resistance 'R' is closed at $t = 0$. The relation between the current I in the circuit and time t is as shown by curve 'a' in the figure. When one or more of parameters E, R & L are changed, the curve 'b' is obtained. The steady state current is same in both the cases. Then it is possible that:



- (A) E & R are kept constant and L is increased.
 (B) E & R are kept constant and L is decreased
 (C) E & R are both halved and L is kept constant
 (D) E & L are kept constant and R is decreased
3. A particle with charge 'q' is travelling with velocity 'v' parallel to a wire with a uniform linear charge distribution λ per unit length. The wire also carries a current I as shown in the fig. The velocity with which particle travels in a straight line parallel to the wire at a distance 'r' away is. (c = Speed of light in vacuum)



(A) $\frac{\lambda}{\mu_0 \epsilon_0 I}$

(B) $\frac{\lambda}{2\mu_0 \epsilon_0 I}$

(C) $\frac{\lambda c^2}{I}$

(D) $\frac{2\lambda}{\mu_0 \epsilon_0 I}$

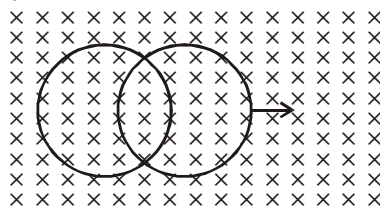


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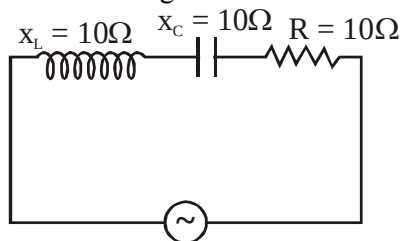
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4. In the diagram shown, a uniform magnetic field is present perpendicular to the plane of the paper. Both the rings are identical and have a constant resistance per unit length. The left ring has been kept fixed at its position and the right ring is slide uniformly on the left ring towards the right hand side. Which of the following statement(s) is/are true ?

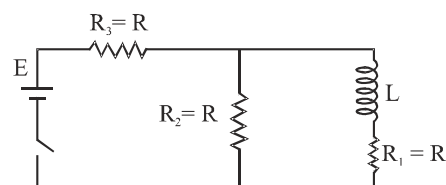


- (A) The emf induced in the left ring is zero
 (B) The emf induced in both the rings is non zero.
 (C) The magnetic force acting on the right ring is zero.
 (D) The magnetic force acting on both the rings is non-zero.
5. A point charge of specific charge $\frac{q}{m} = 0.1 \text{ C/kg}$ is projected in uniform magnetic field. The particle moves in magnetic field such that its position vector at any instant is given by $\vec{r} = 3 \sin t \hat{i} + 3 \cos t \hat{j} + 4t \hat{k}$. Select correct statements from following :
- (A) Magnetic field in space is 10 T
 (B) The distance traveled by the particle in 5 s is 20 m
 (C) Power of magnetic force is zero
 (D) The radius of curvature of the path is 3 m
6. In the given AC circuit, which of the following is/are incorrect :



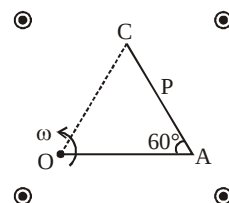
$$\varepsilon = 100 \sin(100\pi t + \pi/2)$$

- (A) Voltage across resistance is lagging by 90° than the voltage across capacitor.
 (B) voltage across capacitor is lagging by 180° than voltage across inductor.
 (C) voltage across inductor is leading by 90° than voltage across resistance.
 (D) Resistance of the circuit is equal to reactance of circuit.
7. A circuit shown in figure contains ideal inductor and battery. Initially current through inductor is zero and the switch is closed at $t = 0$. When potential difference across inductor is 50% of maximum potential difference across it then select correct alternative/s :
- (A) current through battery is $\frac{7}{12} \frac{E}{R}$.
 (B) current through resistance R_2 is $\frac{5}{12} \frac{E}{R}$.
 (C) current through inductor is $\frac{1}{6} \frac{E}{R}$.
 (D) potential difference across R_1 is $\frac{E}{2R}$.



8. A proton is fired from origin with velocity $\vec{v} = v_0 \hat{j} + v_0 \hat{k}$ in a uniform magnetic field $\vec{B} = B_0 \hat{j}$. In the subsequent motion of the proton :
- (A) its z-coordinate can never be negative.
 (B) its x-coordinate can never be positive.
 (C) its x-and z-coordinates cannot be zero at the same time.
 (D) its y-coordinate will be proportional to its time of flight.

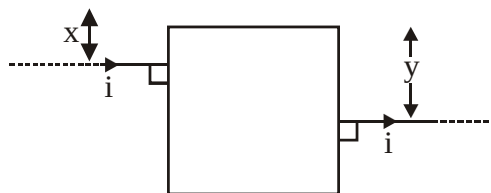
9. A metallic V shaped rod OAC is rotated with respect to one of its end in uniform magnetic field, such that axis of rotation is parallel to the direction of magnetic field. Length of each arm of rod is L and angle between the arms is 60° . P is the mid-point of section AC. Magnitude of magnetic field is B. Then choose the **correct** relations.



- (A) $V_A - V_O = \frac{\omega BL^2}{2}$ (B) $V_A - V_C = \frac{\omega BL^2}{2}$
 (C) $V_C - V_P = \frac{\omega BL^2}{8}$ (D) $V_A - V_P = \frac{\omega BL^2}{8}$

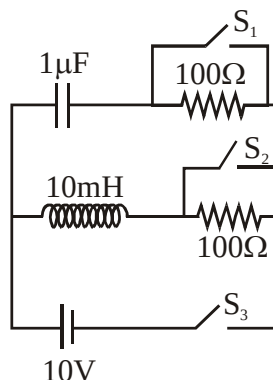
10. A circular loop of conducting wire of radius 1 cm is cut at a point A on its circumference. It is then folded along a diameter through A such that the two semicircular loops lie in two mutually perpendicular planes. In this region a uniform magnetic field $\vec{B}$ of magnitude 100 mT is directed perpendicular to the diameter through A and makes angles of 60° and 30° with the planes of the two semicircles. The magnetic field reduces at a uniform rate from 100 mT to zero in a time interval of 4.28 ms. Therefore :
- (A) instantaneous emf in the two loops are in the ratio $\sqrt{2} : 1$.
 (B) instantaneous emf in the two loops are in the ratio $\sqrt{3} : 1$.
 (C) the total emf between free ends at point A is approximately 5 mV.
 (D) the total emf between free ends at point A is approximately 1.4 mV.

11. A conducting square loop made of homogenous material and having uniform cross section of side 'a' is placed between two semi - infinite wires as shown in figure. Magnetic field intensity at centre of square loop is zero if (Semi - infinite wires and loop lie in same plane).

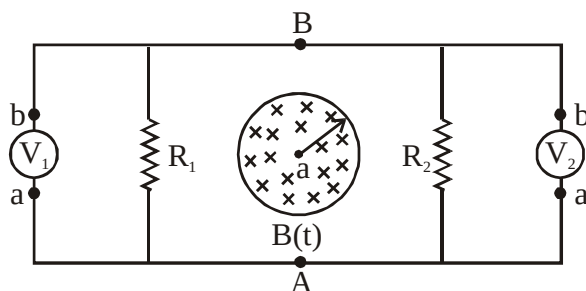


- (A) $x = 0 : y = 0$ (B) $x = 0 : y = a$ (C) $x = \frac{a}{2} : y = \frac{a}{2}$ (D) $x = \frac{a}{4} : y = \frac{3a}{4}$

12. Switches S_1 and S_2 are open while S_3 remains closed for long time such that capacitor becomes fully charged and current in inductor coil becomes maximum. Now, switches S_1 and S_2 are closed and at the same time S_3 is opened at $t = 0$. Assume that battery and inductor coil are ideal. If 'q' denotes the charge on the left plate of the capacitor and $I_{\max}$ denotes the maximum current in the circuit after S_3 is opened, then choose the correct options :

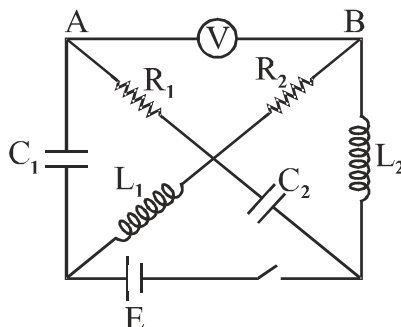


- (A) $q = 10\sqrt{2} \sin \left(10^4 t(s^{-1}) + \frac{3\pi}{4} \right)$ (B) $q = 10\sqrt{2} \sin \left(10^2 t(s^{-1}) + \frac{\pi}{4} \right)$
- (C) $I_{\max} = \frac{1}{10} \text{ A}$ (D) $I_{\max} = \frac{1}{5\sqrt{2}} \text{ A}$
13. When an impedance, consisting of an inductance L and a resistance R in series, is connected across a 12 V, 50 Hz supply, a current of 0.05 A flows which differs in phase from that of the applied potential difference by 60° . If a capacitor of capacitance C connected in series in the above circuit has the effect of bringing the current into phase with the applied potential difference. Select correct alternative :
- (A) $R = 120 \Omega$ (B) $L \approx 0.66 \text{ H}$ (C) $C \approx 15.37 \mu\text{F}$ (D) $C \approx 25.37 \mu\text{F}$
14. The circuit shown in the figure consists of two resistances R_1 & R_2 connected to two ideal voltmeters V_1 & V_2 . A time varying magnetic field $B(t)$ exists in a circular region of radius a and it is directed into the plane of the figure $B(t) = B_0 t$ where B_0 is a positive constant of proper dimensions and t is the time, then choose the correct option(s):

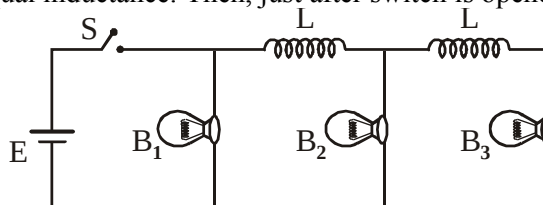


- (A) The emf induced in the circuit is $\pi a^2 B_0$ (B) The reading of V_1 is $\frac{\pi a^2 B_0 R_1}{R_1 + R_2}$
- (C) The reading of V_2 is $\frac{\pi a^2 B_0 R_1}{R_1 + R_2}$ (D) $V_1 = V_2$

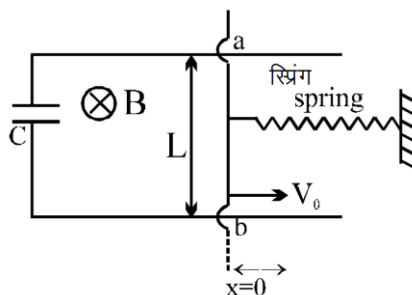
15. For the circuit shown in figure, $R_1 = 3R$, $R_2 = R$, $C_1 = C_2 = C$, and $L_1 = L_2 = L$. The electromotive force of the battery is E . Initially the switch is closed and the system is operating in steady state. The voltmeter is ideal and it measures $V_A - V_B$. The switch is opened at $t = 0$. The reading of voltmeter -



- (A) Just before the opening switch is E (B) Just after the opening of switch is $-2E$
 (C) Long time after opening the switch is E (D) Is zero at any moment
16. In the circuit shown, switch S is closed for a very long time and each bulb is of equal power rating. Both the inductor coil are of equal inductance. Then, just after switch is opened :

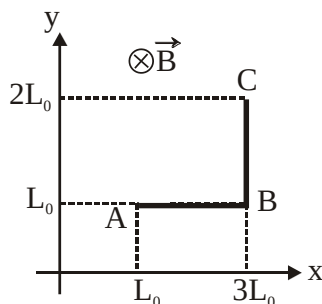


- (A) all the bulbs remain equally bright at all times.
 (B) brightness of bulb B_1 becomes 4 times its original brightness.
 (C) bulb B_1 becomes of zero brightness immediately after opening switch.
 (D) bulb B_2 and B_3 remain equally bright.
17. The conducting connector ab mass m and length L can freely slide on a horizontal long conducting parallel rails connected by capacitor C at one ends (as shown in figure). A non conducting light spring (spring const. K) connected to the connector ab and it is in a relaxed state. The whole system is placed in uniform magnetic field of strength B directed into the plane of rails (as shown in figure) Now at time $t = 0$, connector is suddenly given a velocity V_0 in rightward direction. If resistance and self inductance of circuit is negligible then :

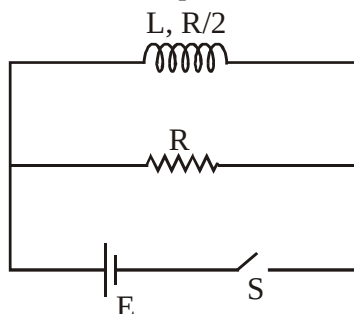


- (A) The acceleration of connector after a displacement of ' x ' is $\frac{Kx}{m}$.
 (B) The acceleration of connector after a displacement of ' x ' is $\frac{Kx}{m - B^2 L^2 C}$.
 (C) The maximum compression in the spring is $V_0 \sqrt{\frac{m}{K}}$.
 (D) The maximum compression in the spring is $V_0 \sqrt{\frac{m + B^2 L^2 C}{K}}$.

18. A L-shaped conducting rod ABC is lying in xy-plane and moving with velocity $\vec{V} = V_0\hat{i} + 3V_0\hat{j}$ as shown in the figure. There exist a non uniform magnetic field which depends on x and y coordinate as $\vec{B} = \frac{B_0}{L_0}(x + y)(-\hat{k})$, where B_0 and L_0 are positive constants with proper dimensions. If $V_P - V_Q$ represents the motional induced emf between the two points P and Q lying on the rod. Then choose the correct option(s) at the instant shown in figure:



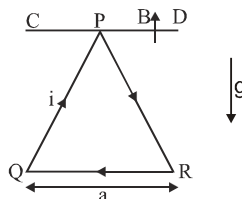
- (A) $V_A - V_B = 18 B_0 L_0 V_0$ (B) $V_B - V_A = 18 B_0 L_0 V_0$
 (C) $V_C - V_A = 27 B_0 L_0 V_0$ (D) $V_A - V_C = \frac{27}{2} B_0 L_0 V_0$
19. A steady state current of 4 A flows in a coil when connected to a 12 V dc source. If the same coil is connected to a 12 V, 50 rad/s ac source a current of 2.4 A flows in the circuit.
 (A) the inductance of the coil is 0.08 H.
 (B) the power developed in the circuit if a 2500 μF capacitor is connected in series with the coil is 17.28 W.
 (C) the inductance of the coil is 0.8 H.
 (D) the power developed in the circuit if a 2500 μF capacitor is connected in series with the coil is 17.28 W.
20. A solenoid of inductance L and resistance $\frac{R}{2}$ is connected in parallel to a resistance R . An ideal battery of emf E is connected across the parallel combination as shown in figure. Switch S is kept closed for long time and it is opened at time $t = 0$. The current through solenoid immediately after switch opened is I and total heat generated in the solenoid after switch opened is H . Then choose the correct option(s):



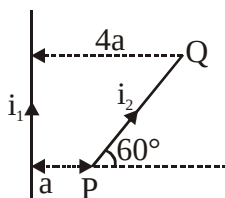
- (A) $I = \frac{E}{R}$ (B) $I = \frac{2E}{R}$ (C) $H = \frac{2}{3} \frac{LE^3}{R^2}$ (D) $H = 2 \frac{LE^3}{R^2}$

INTEGER

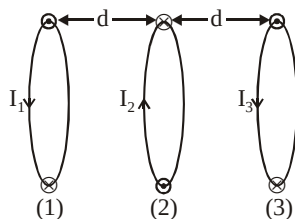
21. A loop PQR formed by three identical uniform conducting rods each of length $a = 20$ cm is suspended from one of its vertices (P) so that it can rotate about horizontal fixed smooth axis CD. Initially plane of loop is in vertical plane. A constant current $i = 10$ A is flowing in the loop. Total mass of the loop is $m = 60$ gm. At $t = 0$, a uniform magnetic field of strength B directed vertically upwards is switched on. If the minimum value of B (in mT) is n , so that the plane of the loop becomes horizontal (even for an instant) during its subsequent motion. Then value of $n/10$ is: ($g = 10 \text{ m/s}^2$)



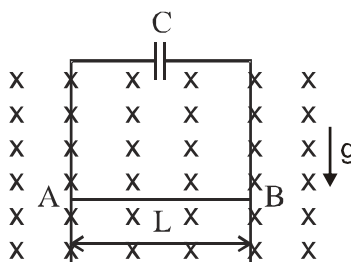
22. A series RLC, AC circuit operates at half its resonant frequency and at a power factor of 0.8. If the value of capacitive reactance X_C is 88Ω then what will be the value of resistance R (in Ω).
23. Consider a long solenoid of radius R which has n turns per unit length. A time dependent current $I = I_0 \sin \omega t$ flows in solenoid, magnitude of electric field at a perpendicular distance $r < R$, from the axis of symmetry of solenoid is found to be $E = \frac{\alpha}{4} \omega \mu_0 n I_0 r \cos \omega t$, where α is pure number and μ_0 is permeability of free space. Find α :
24. A long straight conductor carries current ' i_1 '. A wire PQ carrying current ' i_2 ' is placed as shown. The net force on PQ is $\frac{2\mu_0 i_1 i_2}{\pi} \ell n(x)$, then write the value of ' x '.



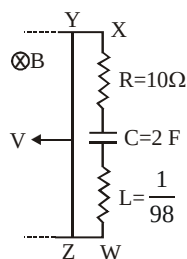
25. Consider three identical conducting loops kept parallel to each other such that their axis coincides. Loop (1), loop (2) and loop (3) self inductance of each loop is L . Current in loop (1) is I_1 , in loop (2) is I_2 and in loop (3) is I_3 , rate of change in current in them at certain instant is $\frac{dI_1}{dt}$, $\frac{dI_2}{dt}$, $\frac{dI_3}{dt}$ respectively. Their mutual inductance is M_{ij} (i.e. for 1 & 2 its M_{12}) current loop (1) and (3) is in same sense and in (2) it is in opposite sense : (Given that if $\frac{dI_1}{dt} = \frac{dI_2}{dt} = \frac{dI_3}{dt}$, emf induced in loop (2) is zero). Find $\frac{L}{M_{12}}$.



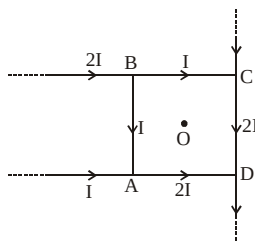
26. A plane spiral coil is made of a thin insulated conducting wire and has 100 turns. Radii of the inside and outside turns are 10 cm and 20 cm respectively. A uniform magnetic field normal to the plane of spiral exist in space. The magnetic field increases at a constant rate 0.3 Tesla/sec. A wire of resistance 1.1Ω is connected between the ends of the spiral. Find the current (in A) induced in the resistance wire to the closest integer. Resistance of the spiral coil is neglected.
27. Two metal bars are fixed vertically and are connected on the top by a capacitor C . A sliding conductor AB of length L slides with its ends in contact with the bars. The arrangement is placed in a uniform horizontal magnetic field directed normal to the plane of the figure. The conductor is released from rest. The displacement is x (in meter) of the conductor after time $t = 2$ sec is:
- [Given $m = 100$ gm, $g = 10$ m/s², $B = 100$ Tesla, $L = 1$ m and $C = 10$ μ F]



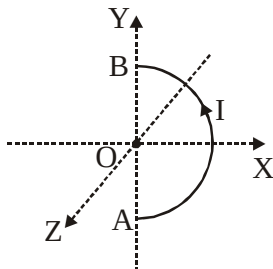
28. In the figure shown magnetic field B is constant and uniform and in the direction perpendicular to the plane of paper. Velocity of a conducting wire YZ is varying with time as $v = v_0 \cos \omega t$. Find the value of ω (in rad/sec.) so that heat generated in circuit will be maximum (neglect friction and resistance of rails)



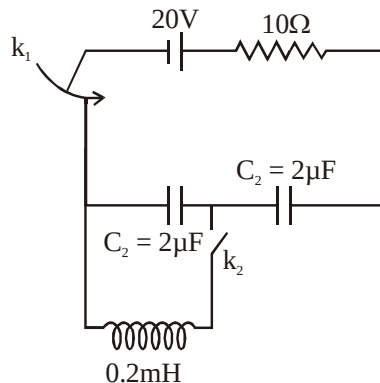
29. When the current in a certain inductor coil is 5.0 A and is increasing at the rate of 10.0 A/s, the magnitude of potential difference across the coil is 140 V. When the current is 5.0 A and decreasing at the rate of 10.0 A/s, the potential difference is 60 V. The self inductance of the coil is x Henry, find the value of x .
30. In the given circuit ABCD is a square of side 'a'. If magnetic induction at point O is $\frac{N\mu_0 I(\sqrt{2}-1)}{\sqrt{2}\pi a}$ find the value of N . Dotted line represents that wire goes upto infinity.



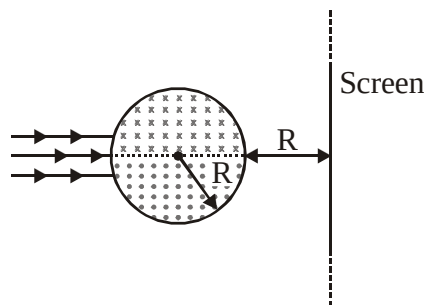
31. A conductor carrying current 60A is in the form of a semicircle AB of radius R and lying in xy -plane with its centre 'O' at origin as shown in the figure. The magnitude of $\oint \vec{B} \cdot d\vec{\ell}$ for the circle $x^2 + z^2 = 3R^2$ in xz -plane due to current in curve AB is $10n\mu_0$. Find the value of n (n is an integer)



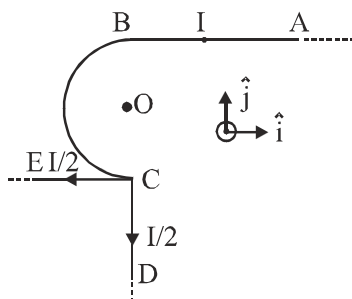
32. In the circuit shown in figure, the key k_1 , was closed for long time. At $t = 0$, key k_1 is opened and k_2 is closed. If the charge on capacitor C_1 is $5\sqrt{N} \mu\text{C}$, at the instant, the energy stored in it is three times of energy stored in inductor. Then value of N is :



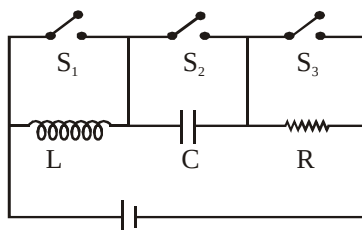
33. Figure shows circular region of radius $R = \sqrt{3} \text{ m}$ in which upper half has uniform magnetic field $\vec{B} = 0.2(-\hat{k}) \text{ T}$ and lower half has uniform magnetic field $\vec{B} = 0.2\hat{k} \text{ T}$. A very thin parallel beam of point charges each having mass $m = 2 \text{ gm}$, speed $v = 0.3 \text{ m/sec}$ and charge $q = +1 \text{ mC}$ are projected along the diameter as shown in figure. A screen is placed perpendicular to initial velocity of charges as shown. If the distance between the point on screen where charges will strike is $4X$ meters, then calculate X .



34. Current I flowing through the semi circular arc BC of radius r splits into two equal parts at the point C as shown in the figure. Net magnetic field at the centre O of the circle are is $\frac{\mu_0 I}{4r} \left(\frac{x}{7} + \frac{1}{2\pi} \right) (+\hat{k})$. Find the value of x .

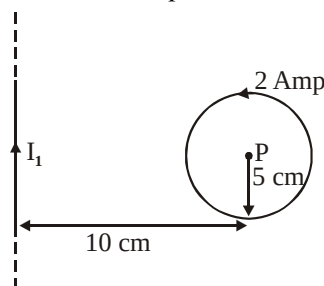


35. Consider the circuit shown in figure. With switch S_1 closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch S_2 closed and the other two switches open, the circuit has a time constant 2 sec. With switch S_3 closed and the other two switches open, the circuit oscillates with a period T . Find T (in sec). (Take $\pi^2 = 10$)

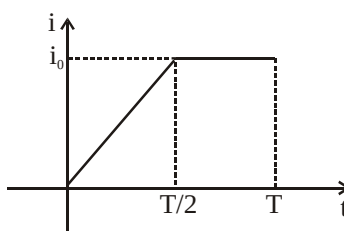


36. An electric bulb is designed to consume 55 W when operated at 110 volts. It is connected to a 220 V, 50 Hz line through a choke coil in series. The inductance (in H) of the coil in series for which the bulb gets correct voltage is X . Find the value of X . (Take $\pi = \frac{22}{7}$)

37. Magnetic field at point P is zero. If current I_1 is equal to 'X' A. Then Find the value of X.

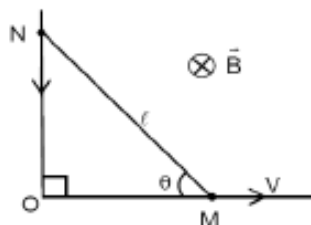


38. One cycle of an alternating current is shown in the graph. R.M.S. value of current from $t = 0$ to $t = T$ is $\sqrt{\frac{2}{x}} i_0$. Find x .



39. The current density $\vec{J}$ inside a long, solid, cylindrical wire of radius $a = 12$ mm is in the direction of the central axis, and its magnitude varies linearly with radial distance r from the axis according to $J = \frac{J_0 r}{a}$, where $J_0 = \frac{10^5}{4\pi}$ A/m². Find the magnitude of the magnetic field at $r = \frac{a}{2}$ in μ T.

40. Figure shows a movable, rigid and conducting connector MN between a fixed L-shaped conducting wire. The whole system is in horizontal plane having vertical magnetic field B and point M is pulled horizontally with constant speed V starting from $t = 0$ by some external agent. Initially, at $t = 0$, if $\theta = 90^\circ$, induced current in the loop OMN is found to be zero at time $t = \frac{l}{\sqrt{K} \times V}$. Find K (Given that M and N are always in contact with the L-shaped wire).



SCQ

Answer Q.41 Q.42 and Q.43 by appropriately matching the information given in the three columns of the following table.

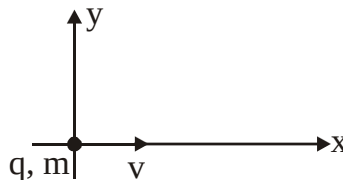
A charged particle (electron or proton) is introduced at the origin ($x = 0, y = 0, z = 0$) with a given initial $\vec{v}$. A uniform electric field $\vec{E}$ and a uniform magnetic field $\vec{B}$ exist everywhere. The velocity $\vec{v}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in columns I, II and III respectively. The quantities E_0 and B_0 are positive in magnitude.

Column -I	Column -II	Column -III
(I) Electron with $\vec{v} = \frac{E_0}{B_0} \hat{y}$	(i) $\vec{E} = E_0 \hat{z}$	(P) $\vec{B} = -B_0 \hat{x}$
(II) Proton with $\vec{v} = 0$	(ii) $\vec{E} = -E_0 \hat{y}$	(Q) $\vec{B} = B_0 \hat{z}$
(III) Proton with $\vec{v} = 2 \frac{E_0}{B_0} \hat{x}$	(iii) $\vec{E} = -E_0 \hat{x}$	(R) $\vec{B} = B_0 \hat{y}$

41. In which case will the particle move in a straight line with constant velocity ?
 (A) (III) (i) (Q) (B) (II) (ii) (R) (C) (II) (iii) (P) (D) (I) (iii) (Q)
42. In which case would the particle move in a straight line along the negative direction of y-axis (i.e., move along $-\hat{y}$) ?
 (A) (II) (ii) (P) (B) (II) (ii) (R) (C) (III) (ii) (Q) (D) (I) (iii) (R)
43. In which case will the particle describe a helical path with axis along the positive z-direction ?
 (A) (II) (iii) (P) (B) (I) (ii) (R) (C) (III) (ii) (R) (D) (III) (i) (Q)

Paragraph for Question Nos. 44 to 45

At $t = 0$ a charged particle of mass m , charge q moving with velocity $\vec{V} = V \hat{i}$ is at origin. Either electric field $\vec{E} = E_0 \hat{j}$ or magnetic field $\vec{B} = -B_0 \hat{k}$ is present in the region. For $t > 0$, E and B are switched on in following manner.



E	B	Time
0	$\vec{B}$	$0 \leq t \leq t_0$
$\vec{E}$	0	$t_0 < t \leq 2t_0$

Where $t_0 = \frac{\pi m}{2qB_0}$

and numerically $VB_0 = \pi E_0$

44. The speed of particle at $t = 2t_0$ will be :

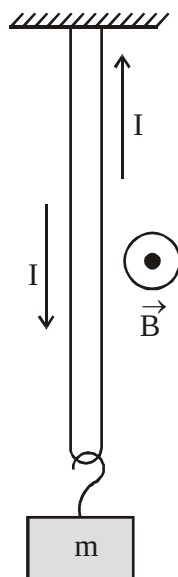
(A) $\sqrt{2}V$ (B) $\frac{V}{\sqrt{2}}$ (C) V (D) $\frac{3V}{2}$

45. The coordinate of the particle at $t = 2t_0$ will be, if $R = \frac{mV}{qB}$:

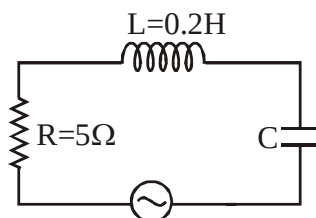
(A) $x = R; y = R\left(1 + \frac{5\pi}{8}\right)$ (B) $x = -R; y = R\left(1 + \frac{5\pi}{4}\right)$
 (C) $x = 2R; y = R\left(1 + \frac{5\pi}{4}\right)$ (D) $x = R\sqrt{2}; y = R\sqrt{2}\left(1 + \frac{5\pi}{8}\right)$

Paragraph for Questions 46 and 47

Consider a loop of freely deformable conducting wire with thermal insulation of length 2ℓ , the two ends of which are fixed (permanently) to the ceiling. A load of mass m is fixed to the middle of the wire (the mass of the wire is negligible). There is also a horizontal magnetic field of induction B ; free fall acceleration is g . A current I is passed through the wire. Neglect the field induced by the wires.

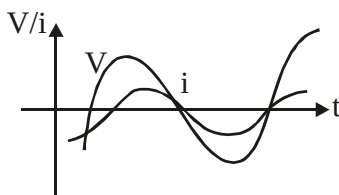


46. If the block is in equilibrium then select correct alternatives.
- (A) The left and right part of wire will take shape of circular arc.
 (B) The left and right part of wire will take shape of parabolic arc.
 (C) Tension is constant along the wire.
 (D) Total magnetic force on the wire must be equal to weight of the load.
47. Select correct alternative(s) for equilibrium condition.
- (A) The maximum height by which the load can be lifted is $\ell \left(1 - \frac{2}{\pi}\right)$.
 (B) The maximum height by which the load can be lifted is $\ell \left(1 - \frac{3}{\pi}\right)$.
 (C) To lift the load by $\ell \left(1 - \frac{3}{\pi}\right)$ the current in wire should be $\frac{mg\pi}{3\sqrt{3}\ell B}$.
 (D) To lift the load by $\ell \left(1 - \frac{3}{\pi}\right)$ the current in wire should be $\frac{mg\pi}{\sqrt{3}\ell B}$.
48. An alternating voltage $E = 6 \sin 20t + 8 \cos 20t$ is applied to a series resonant circuit as shown. The incorrect statement is:



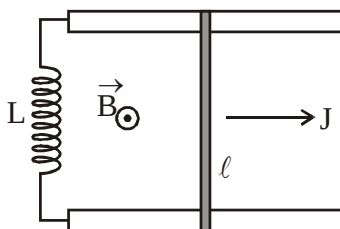
- (A) The capacitance C is 12.5mF .
 (B) The resonant current in the circuit is $\sqrt{2}\text{A}$.
 (C) Average power dissipated in the circuit is 6 W .
 (D) The quality factor of the current is 0.8 .

49. Graph shows variation of source emf V and current i in a series RLC circuit, with time.



Choose the correct option (s) :

- (A) The current leads the emf in the circuit.
 (B) The circuit is more inductive than capacitive.
 (C) To increase the rate at which energy is transferred to the resistive load, L should be decreased.
 (D) To increase the rate at which energy is transferred to the resistance load, X_C should be increased.
50. Two parallel resistanceless rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ , mass m and negligible resistance is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the correct option (s).



- (A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor

$$d = \sqrt{\frac{3J^2 L}{4B^2 \ell^2 m}}.$$

- (B) Current flowing through the inductor at the instant when velocity of the conductor is half of the

initial velocity is $i = \sqrt{\frac{3J^2}{2Lm}}.$

- (C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor

$$d = \sqrt{\frac{3J^2 L}{B^2 \ell^2 m}}.$$

- (D) Current flowing through the inductor at the instant when velocity of the conductor is half of the

initial velocity is $i = \sqrt{\frac{3J^2}{Lm}}.$

ANSWER KEY TO DPP # 03 (JEE-ADVANCED)

1. (A,B,D)	2. (A,C)	3. (A,C)	4. (A,D)	5. (A,C)
6. (A,D)	7. (A,B,C)	8. (B,D)	9. (A,C,D)	10. (B,C)
11. (B,C,D)	12. (A,D)	13. (A,B,C)	14. (A,B)	15. (A,B)
16. (B,D)	17. (D)	18. (A,D)	19. (A,D)	20. (B,C)
21. 40	22. 88	23. 02	24. 02	25. 02
26. 02	27. 10	28. 07	29. 04	30. 03
31. 03	32. 12	33. 03	34. 07	35. 02
36. 01.21	37. 12.57	38. 03	39. 10	40. 02
41. (D)	42. (B)	43. (D)	44. (D)	45. (A)
46. (A,C)	47. (A,C)	48. (C)	49. (B,C,D)	50. (A)